IT5006 TEAM 9

Predicting Delivery Lead Time and Low-Review Risk on Olist

Phase II — Modelling and Evaluation

Regression and classification results

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Phase II Report | Model development, evaluation and implications

1. Introduction and Problem Definition

E-commerce platforms face a trade-off between attractive delivery promises and reliable fulfilment. Faster promises can increase sales, but overly aggressive commitments may undermine retention (Cui et al., 2024). Late deliveries are also associated with lower review ratings and customer satisfaction (Ravula, 2023; Vakulenko et al., 2019). Building on these findings, this project uses Olist data to predict delivery lead time and the risk of a 1–2-star review. It compares models against simple baselines to assess their potential to support fulfilment planning and prioritised customer outreach.

1.1 Regression: predicting delivery lead time

Problem and target. The regression question is: How many days will elapse between an order being placed and its receipt? The continuous response is (order_delivered_customer_date − order_purchase_timestamp) in fractional days. Predictions are generated at purchase from information assumed available then, including order composition, seller and route, timing and the delivery promise. The modelling population contains 96,470 orders with completed deliveries and valid duration targets.

Stakeholders and use. Olist fulfillment planners and customer-service teams can use estimated lead times to screen potentially slow routes, check fulfilment capacity and communicate provisional arrival expectations. Success is assessed primarily by prediction error in days. An estimate is not a guaranteed delivery date, particularly for long-tail delays.

1.2 Classification: predicting low-review risk

Problem and target. The classifier predicts whether an order receives a 1–2-star review rather than 3–5 stars, retaining the earliest review per order. The prediction point is reconstructed as T = min(order_delivered_customer_date, order_estimated_delivery_date), representing delivery or the promised date for an undelivered parcel. This is a proxy for the survey trigger, not a logged intervention time. Predictors use only information available by T; later delivery events and review content are excluded. The cohort contains 97,490 reviewed orders with items, purchased September 2016–August 2018, after excluding 426 reviews answered before T.

Stakeholders and use. Olist customer-experience and seller-quality teams can rank orders by risk to prioritise factual shipping updates, service recovery and seller review within a defined outreach capacity. Performance must reflect ranking of the minority low-review class (14.15%), coverage of dissatisfied customers and unnecessary contacts. Prediction does not demonstrate that contacting an order will prevent a negative rating.

2. Methodology

2.1 Data preprocessing and analytical population

Regression. Seven Olist tables—orders, order items, customers, sellers, products, geolocation and category translations—were linked through documented identifiers. Item records were aggregated before joining to preserve one observation per order and avoid multiplying basket totals. Of 99,441 source orders, 96,478 were marked delivered; eight without customer-receipt timestamps were excluded. This produced 96,470 orders with valid non-negative purchase-to-receipt durations. Long but valid lead times were preserved, including 306 deliveries exceeding 60 days. Geographic cleaning removed 29 out-of-range coordinates and calculated postal-prefix medians from unique locations. Missing route distance (477 orders) and product weight (22 orders) did not trigger

wholesale deletion; 84 chronology-flagged records were retained with a separate sensitivity check.

Classification. Nine source tables additionally included payments and reviews. Item and payment transactions were first consolidated to the order level; if several reviews were present, the earliest review was retained after sorting by review creation and answer timestamps. The in-window feature table contained 99,421 orders, and 97,916 had an observed review and at least one item. Excluding 426 reviews submitted before the intended prediction time T left 97,490 eligible orders, including 13,796 low reviews (14.15%). This population includes parcels that had not arrived by T, unlike the completed-delivery regression population. The response label is never imputed. In the classification cohort, subsequent delivery-duration measures are unavailable for approximately 10% of orders; main category and route distance are missing for approximately 1.4% and 0.5%.

Both problems are defined per order and fit model-specific preprocessing within training pipelines. Imputation, missingness indicators, encoding, scaling and classification history construction are fitted only on the relevant training partition or fold. CatBoost supports native missing numerical values. Different outcomes and prediction points explain the distinct cohorts.

2.2 Data splitting and leakage prevention

Regression. A reproducible random 67/33 split (seed 33) produced 64,634 training and 31,836 test orders. Candidate models and preprocessing were evaluated with five shuffled cross-validation folds on training data; the fixed test partition supplies the recorded single-model performance assessment. Scaling, imputation, categorical transformations and the learned probability of delivery exceeding 30 days were trained within fitting partitions. The auxiliary slow-delivery score was cross-fitted for training observations to prevent their own delivery targets entering the generated feature.

Classification. Stratified 67/33 train/test and 80/20 train/validation splits (seed 5006) produced 52,254 training, 13,064 validation and 32,172 test orders, each with approximately 14.15% low reviews. Five shuffled stratified training folds supported tuning and out-of-fold predictions. Validation PR-AUC determined model and ensemble selection; validation F1 determined thresholds, which were frozen for testing. Five imbalance strategies were compared within training pipelines.

Every predictor is restricted to what is observable at T (Section 1.2); later delivery events and the review itself are excluded by the feature-role audit. A fold-fitted history transformer rebuilds seller and route outcome statistics using only the training rows of each fold and events occurring before the target order's purchase. Seller-review history dates the outcome at review_answer_timestamp, not the survey-creation date. Identifiers and historical-construction context—including raw ratings, delivery outcomes and event timestamps—are removed before estimator fitting. The same isolation applies to stacking and sampling pipelines.

2.3 Feature engineering and rationale

Appendix A (Table A1b and the linked feature catalogue) and Appendix B (Table B1b) give each input's rationale and prediction point; this section gives the reasoning by group.

Regression. Quoted promise — the retailer's own forecast and the expectation the customer judges against; Cui et al. (2024) show promised speed itself shapes behaviour, making it a candidate predictor and screening reference. Geography and route — distance, state and city routes, cross-state flags, plus promise × distance interactions, because last-mile transit dominates delivery-time variation (Vakulenko et al., 2019). Order composition, weight and volume — multi-seller baskets need coordinated

shipments; heavy items follow different carrier classes. Calendar — cyclical month, weekday and hour plus elapsed time, to absorb demand peaks and drift. Seller identity — the primary seller's own record, pooled below 20 orders. Learned risk — risk_p30, a cross-fitted purchase-time probability of exceeding 30 days, with a small marginal regression gain.

Classification. Literature-derived inputs include order and payment value, freight ratio, working-day delivery duration and promise deviation (Zaghloul et al., 2024), calculated as of T. Calendar-day counterparts remain available; their importance rankings need not agree. Delivery status and progress represent the lateness signal identified in Phase 1. Unobserved delivery measures remain missing with indicators. Basket composition represents shipment complexity, while geography, timing and payment describe route exposure and order value (Akturk et al., 2022). Seller and route history use earlier training-fold outcomes, smoothing with pseudo-count 20 and pooling below 50 orders. Review text is excluded.

2.4 Models and baselines

Regression. Ordinary least squares and a single decision tree serve as simple linear and nonlinear baselines. Ridge regularizes linear coefficients; Random Forest aggregates bootstrapped trees; histogram-based gradient boosting (HGB) learns sequential nonlinear corrections. The Random Forest reported in the main comparison uses F1 inputs and an untransformed target. A separate variant with a log-transformed target and additional time and cross-fitted risk features was evaluated in the exploratory stacking comparisons.HGB uses absolute-error loss to align training with MAE. A matched 31-leaf HGB control tests whether the tuned model earns its complexity, and constrained stacking of Ridge, Random Forest and HGB (with an ExtraTrees variant, Appendix A3) tests whether combining families adds anything.

Classification. Three families were compared: (A) baseline logistic regression; (B) a single decision tree, Random Forest and CatBoost; and (C) probability averaging and logistic stacking of linear and tree models. The majority-class predictor and the rule "flag orders not delivered by T" provide no-skill and operational reference points. Logistic regression gives an interpretable linear comparator; the tree ensembles capture nonlinear interactions between promises, delivery progress and order characteristics.

2.5 Training, tuning and cross-validation

Regression. Candidate feature packages and bounded hyperparameter configurations were evaluated by mean five-fold cross-validated MAE. The final HGB search compared eight settings around a common specification. The selected model has absolute-error loss, 300 boosting iterations, learning rate 0.05, 63 maximum leaf nodes, 30 minimum observations per leaf and L2 regularization 10; early stopping was disabled. It was refitted on all 64,634 training observations before testing. Ensembles were compared on the same five outer folds, with three inner folds and seed 33. The historical stack tuned Ridge and RF in the inner folds. The current comparison replaces RF with the fixed log-target time/risk configuration while retaining the earlier selected Ridge and matched HGB predictions. Combination weights minimize MAE on inner out-of-fold predictions, subject to nonnegative weights summing to one and zero intercept. The risk feature is cross-fitted within each regression fitting partition. No final full-training ensemble or ensemble test score was produced.

Classification. RandomizedSearchCV evaluated up to 30 parameter draws per searched family, with five-fold stratified CV and average precision as the optimization target. The chosen logistic model used C = 0.1; Decision Tree depth 9 and minimum leaf size 138; Random Forest 478 trees, depth 21, minimum leaf size 70 and maximum feature fraction 0.356; CatBoost depth 4, 1,000 iterations and learning rate approximately

0.0318 (Appendix Table B2a). CatBoost early stopping was also evaluated against the searched model. Preprocessing uses median imputation, missing indicators, log-transformed or scaled linear numerics and categorical encoding, while the CatBoost pipeline supports its categorical representation. All fitted steps—including history construction—are repeated inside the CV.

For classification ensembles, logistic regression, Random Forest and CatBoost provide five-fold out-of-fold probabilities for arithmetic means, a convex weighted average and a logistic meta-model. The reported stacking scores come from the probability-based stack in baseline_variant_review.ipynb. The supplementary ensemble_comparison.ipynb also evaluates a logit-based stack using the same selected base-model configurations. Validation PR-AUC determines whether any ensemble improves on the best single learner by at least 1% relatively. The class-imbalance study separately compares no weighting, class weights, random oversampling, SMOTE and ADASYN; resampling is confined to fitting data. This separates improvements in ranking from shifts in the probability scale or selected cutoff.

2.6 Evaluation metrics

Regression. MAE, reported in days, is the primary selection criterion and reflects average absolute forecast error. RMSE gives greater weight to unusually large errors, while R^2 compares predictions with a constant-mean reference. Mean signed error (prediction minus observation), the 90th percentile absolute error and the share within ±3 days diagnose underprediction and operational reliability. Errors are further examined by route and observed lead-time group, including unusually slow deliveries. Model importance is evaluated by the rise in MAE when groups of predictors are permuted.

Classification. Average precision (AP, reported as PR-AUC) is the selection metric; its no-skill level is the 14.15% low-review prevalence. ROC-AUC supplements ranking assessment. Table 2 reports weighted precision, recall and F1 following Zaghloul et al. (2024); weighted recall equals accuracy for this single-label task. These averages are dominated by the majority class, so positive-class precision and recall are also reported. Thresholds maximise validation positive-class F1 and remain fixed on test. Brier score measures probability prediction error, reflecting both calibration and discrimination; imbalance corrections can affect these differently (van den Goorbergh et al., 2022).

3. Results and Model Interpretation

3.1 Regression: model selection and test performance

Models were developed from simple family baselines, with selection based on five-fold training CV MAE (seed 33). Appendix A2 documents model configurations and the bounded tuning search, while Appendix A3 reports training, cross-validation and exploratory stacking results. Table 1 presents test performance. Ridge improved on OLS (CV MAE 4.9836 vs 5.0196 days; test 4.9214 vs 4.9607), while RF with F1 inputs and an untransformed target improved on the single tree (CV 4.7269 vs 5.0585; test 4.6525 vs 5.0297). These comparisons involve both model and feature changes. On identical inputs, the selected 63-leaf HGB improved on the 31-leaf control (CV 4.2378 vs 4.2729; test 4.1505 vs 4.1923).

Separate exploratory stacking experiments used an enhanced RF with time/risk features and a log-transformed target. Despite its lower standalone CV MAE of 4.4152 days, the constrained three-model stack assigned HGB 100% weight in every fold. The HGB–ExtraTrees stack improved MAE by only 0.0013 days (0.03%). These small gains did not justify the additional complexity, so standalone HGB was retained. Appendix A3

documents the stacking comparisons; no final ensemble refit or test evaluation was performed.

Table 1. Regression test performance (n = 31,836)

Model	Family step	MAE (days)	RMSE (days)	R^2
OLS — F0	Linear, baseline	4.9607	7.8947	0.2926
Decision tree — F0	Tree baseline	5.0297	8.0311	0.2680
Ridge — F1, log inputs	Linear, regularised	4.9214	7.8537	0.3000
Random forest — F1, original target	Tree ensemble	4.6525	7.5879	0.3466
HGB — 31 leaves, time/risk	Boosting, control	4.1923	7.5028	0.3611
HGB — 63 leaves ,time/risk(selected)	**Boosting, tuned**	**4.1505**	**7.4651**	**0.3675**

Against non-model references the gain is larger: using the delivery promise itself as the forecast gives MAE 12.72 days and the training median 6.08, so HGB removes two-thirds of the promise's error. The error profile still matters for use. Mean signed error is −1.33 days, so the model underpredicts on average; only 56.9% of orders fall within ±3 days and the 90th-percentile absolute error is 9.1 days. Errors are route-dependent — MAE 4.82 days cross-state against 2.95 within-state — and larger in the tail: orders taking over 60 days (n = 99) have MAE 66.2 days, indicating that the current model does not reliably predict these extreme delays from the available purchase-time features.

3.2 Classification: model selection and test performance

Table 2 includes two reference benchmarks: The majority class (predict no low reviews; PR-AUC equals the 14.15% base rate) and the delivery-status rule, which flags every order undelivered at the promised date — a policy usable without a model, contacting 9.9% of orders at precision 0.59 and recall 0.41.

Models were selected on validation PR-AUC (Appendix Table B2b). Regularising logistic regression produced only a small change (0.468 to 0.469). Random Forest reached 0.491, exceeding the single tree (0.469), CatBoost (0.483) and stacking (0.489). Stacking therefore failed the required 1% relative validation gain and was not retained. Its test PR-AUC of 0.508 versus 0.505 for RF is descriptive; no paired uncertainty analysis establishes an advantage. Validation scores are below CV means, but rankings are not identical: logistic regression exceeds the tree in CV, while their validation order reverses. These comparisons do not establish a uniform shift in difficulty across models.

Model	CV PR-AUC ± SD	W-Precision	W-Recall	W-F1	ROC-AUC	PR-AUC
Majority baseline	—	0.737	0.858	0.793	0.500	0.142

Model	CV PR-AUC ± SD	W-Precision	W-Recall	W-F1	ROC-AUC	PR-AUC
Delivery-status rule	—	0.862	0.876	0.867	0.682	0.326
Logistic Regression	0.499 ± 0.013	0.865	0.870	0.867	0.786	0.490
Decision Tree	0.490 ± 0.008	0.859	0.848	0.853	0.771	0.484
Random Forest (selected)	**0.515 ± 0.010**	**0.863**	**0.861**	**0.862**	**0.788**	**0.505**
CatBoost	0.507 ± 0.010	0.864	0.867	0.866	0.789	0.500
Stacking	—	0.866	0.872	0.868	0.790	0.508

Table 2. Classification test performance (n = 32,172; 14.15% low reviews)

W denotes support-weighted metrics across both classes; weighted recall equals accuracy. These averages are dominated by the 85.85% majority class and do not replace positive-class metrics. Selection uses validation PR-AUC (Appendix B2).

The forest (478 trees, depth 21, minimum leaf 70, max_features 0.356) scores test PR-AUC 0.505, 3.6× no-skill and 0.18 above the rule. At the frozen threshold (0.590) it flags 14.6% of orders: 2,381 true positives, 2,304 false positives, 2,172 missed — precision 0.51, recall 0.52 (Figure B3b); the top 10% of risks capture 42.8% of low reviews. Detection is concentrated where the signal is: recall is 0.99 among the 3,186 orders undelivered at T (59% low-review rate) but 0.19 among the 28,986 delivered by T (9.2%; PR-AUC 0.227). The current model remains less effective at detecting low reviews among orders delivered by T; these results do not establish an upper limit on what the available features can predict.

Removing RF class weights gives similar aggregate ranking performance (test PR-AUC 0.5058 versus 0.5047) and lower Brier error (0.0923 versus 0.1453), not proof of identical rankings or perfect calibration (Table B3a). Random oversampling, SMOTE and ADASYN did not improve RF validation PR-AUC. The main model and operating points remain the class-weighted RF; the unweighted refit is a sensitivity comparison and a candidate for further calibration assessment before displaying probabilities.

3.3 Interpreting the different predictions

Feature importance reflects the prediction stage. Regression relies on purchase-time geography, calendar and promise information. Grouped permutation importance (Appendix Figure A5a) puts route geography first (+1.35 days MAE when shuffled), purchase calendar second (+0.98) and the quoted promise third (+0.53); price, category and weight add under 0.05 days each. These results motivate route-specific promise and carrier review, and monitoring for calendar drift. These values describe model reliance, not causal effects.

For the class-weighted RF, validation permutation importance (Appendix Figure B5a) ranks item count first (PR-AUC decrease 0.030), then delivered-by-T status (0.018), working-day promise deviation and carrier transit days (about 0.012 each; Figure 2). SHAP on 1,500 validation rows in the same class-weighted forest also ranks number of

items and delivered-by-T first, but only four of its top six features overlap with the permutation ranking; seller identity and actual delivery time enter the SHAP top six. These methods measure different aspects of model reliance. The results motivate testing basket-status updates and seller review, while weaker detection among delivered orders limits outreach coverage. Neither importance method establishes causality or the benefit of an intervention.

4. Stakeholder Implications and Limitations

Together, purchase-time duration estimates and risk scores at T could support delivery-expectation management and targeted service recovery. Their operational value requires prospective evaluation.

4.1 Regression: fulfilment and expectation management

Operations teams could use purchase-time estimates to screen exposed routes and complex orders for fulfilment checks. An MAE near four days is not a delivery guarantee. Persistent underprediction of long delays calls for escalation rules or uncertainty ranges for unusual routes.

4.2 Classification: prioritised service recovery

Customer-experience teams could rank orders at T and choose an outreach quota based on validation performance, staffing and the precision–recall trade-off. As a retrospective ranking diagnostic, the highest-risk 10% of test orders captured 42.8% of low-review cases. Expanding the queue increases workload and may include more orders that would not receive low reviews. A controlled pilot should assess whether shipment checks, seller contact and factual updates improve customer outcomes.

4.3 Limitations and Constraints

Random splits do not establish performance on future months or new sellers. Regression test results preceded later RF/HGB tuning and stacking development; training CV guided those stages, but the test set no longer independently confirms the full selection process. This does not establish direct tuning on test scores. Classification records document validation-based choices without proving that test results were inspected only once.

Regression excludes incomplete deliveries and retains 84 chronology-flagged records; excluding 25 flagged test orders changes MAE from 4.1505 to 4.1523 without refitting. Classification covers reviewed orders, risking response-selection bias. Fold-fitted history protects the information boundary, but calibration and drift still require prospective checks.

The classifier exceeds twice the no-skill PR-AUC (0.505 vs 0.142, 3.6×). A validation-derived 20% capacity cutoff yields test recall 0.605 and precision about 0.418 (Appendix B4a), meeting the operating target retrospectively; the F1 cutoff does not. As in Section 1.2, T approximates the survey trigger from two timestamps rather than a logged send event. For 2.7% of cohort orders the review was created before T, so T falls slightly after the actual trigger. This observed mismatch rate is not a timing-error bound. Prospective timing and calibration checks remain necessary.

5. Conclusion

The selected 63-leaf HGB achieves test MAE 4.1505 days; negligible ensemble gains support retaining it, although extreme delays remain difficult. The selected RF achieves PR-AUC 0.505 and positive-class recall 0.523 at its frozen F1 threshold. Both support

operational pilots, subject to the evaluation limitations above and prospective measurement of workload, calibration and intervention outcomes.

References

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GITHUB REPOSITORY

https://github.com/IT5006-Group9/IT5006-Group9-Olist-Project

AI Tool Declaration

We used GPT-5, Claude Fable 5.1 to generate and validate ideas, improve expression and refine our writing. We also used the models to debug our code. We are responsible for the content and quality of the submitted work.

Appendix A. Delivery Lead-time Regression supporting evidence

A1 Data preparation and feature specification

Table A1a. Cohort and data-quality checks

Check	Orders
Source orders	99,441
Orders marked delivered	96,478
Delivered orders missing receipt timestamps; excluded	8
Eligible completed orders with non-negative targets and items	96,470
Observed durations exceeding 60 days; retained	306
Missing maximum distance / average product weight; retained	477 / 22
Chronology-flagged orders; retained in main analysis	84 (59 train; 25 test)
Training / test orders	64,634 / 31,836

The seven source tables were orders, items, customers, sellers, products, geolocation and category translations. Item aggregation and many-to-one joins maintained one row per order. Geographic screening excluded 29 coordinate records outside latitude −35° to 6° or longitude −75° to −25°; postal-prefix coordinates were medians of unique coordinate pairs.

Table A1b. Feature designs used in the comparisons

Design	Definition
F0	16 baseline fields covering the promise, distance, price/freight, item/seller counts, average weight, missingness, customer state, route/category groups and purchase month/weekday/hour.
F1	F0 plus six cyclical calendar variables, three promise/distance/cross-state interactions and one state-route field.
Expanded purchase inputs	F1 plus primary seller identity/composition, detailed routes, product composition, weight and volume.
Time and risk	Elapsed purchase days, purchase year-month, and a cross-fitted probability of delivery exceeding 30 days.

Archived purchase attributes are assumed available at purchase. The final fitted matrix has 1,042 encoded columns, not 1,042 original variables; fold-specific category vocabularies may differ. Learned preprocessing and risk construction use fitting partitions only.

Full feature catalogue and availability assumptions

Source-table inventory

A2 Model configurations and bounded tuning

Table A2a. Configurations of the six models reported in the main tables

Model	Inputs	Key settings
OLS	F0	Ordinary least squares; untransformed target.

Single tree	F0	Maximum depth 8; minimum leaf size 20; seed 33.
Ridge	F1; log inputs	α = 100; seven selected inputs transformed with log1p; untransformed target.
Random forest	F1; untransformed target	100 trees; maximum depth 24; minimum leaf size 10; max_features = 0.8; bootstrap enabled; max_samples = 0.8; seed 33; untransformed target.
HGB control	Expanded + time/risk	Absolute-error loss; 300 iterations; learning rate 0.05; 31 leaves; minimum leaf size 30; L2 = 10.
Selected HGB	Expanded + time/risk	Same settings as the HGB control, except 63 leaves.

HGB used seed 33 with early stopping disabled. Ridge log-transformed distance, price, freight, freight ratio, item count, seller count and average product weight. Numerical imputation, encoding and applicable scaling were fitted within the relevant training partitions.

Table A2b. HGB parameter comparison on the time-plus-risk inputs

Change from control	CV MAE ± SD (days)
31-leaf control	4.2729 ± 0.0408
15 leaves	4.3247 ± 0.0437
63 leaves (selected)	4.2378 ± 0.0369
Min. leaf size 15	4.2741 ± 0.0399
Min. leaf size 60	4.2709 ± 0.0420
L2 = 0	4.2754 ± 0.0383
600 iterations	4.2393 ± 0.0397
Learning rate 0.10	4.2418 ± 0.0390

Each candidate changed one setting from the control. All eight settings were evaluated using five folds; the selected 63-leaf configuration was additionally checked without the risk feature (CV MAE 4.2399 ± 0.0477 days). This was a bounded search, not a full Cartesian grid.

The auxiliary HGB classifier used 200 iterations, learning rate 0.05, 15 leaves, minimum leaf size 40, L2 = 10 and seed 33, with early stopping disabled. Each outer regression fit generated training risks through three-fold cross-fitting; final full-training risks used five-fold out-of-fold predictions.

Complete model specifications

A3 Fold results and exploratory stacking

Table A3a. Regression training and cross-validation performance

Model	Role or family	CV MAE, mean ±SD(days)	Training MAE (days)
OLS – F0	Linear baseline	5.0196 ± 0.0423	5.0045

Model	Role or family	CV MAE, mean ±SD(days)	Training MAE (days)
Single Decision Tree –F0	Tree baseline	5.0585 ± 0.0532	4.9751
Ridge – F1, log inputs	Linear variant	4.9836 ± 0.0412	4.9724
Random Forest – F1,original target	Tree ensemble	4.7269 ± 0.0532	3.8850
HGB: 31 leaves,time/risk	HGB control	4.2729 ± 0.0408	4.1118
HGB: 63 leaves,time/risk	Final model	4.2378 ± 0.0369	3.9295

Table A3b. Held-out MAE for the six reported models across five folds (days)

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
OLS — F0	5.0100	5.0680	5.0594	4.9862	4.9746
Single tree — F0	5.0398	5.1033	5.1260	5.0123	5.0109
Ridge — F1, log inputs	4.9651	5.0337	5.0222	4.9521	4.9452
RF — F1, original target	4.7366	4.7724	4.7819	4.6656	4.6781
HGB — 31 leaves, time/risk	4.2762	4.2981	4.3189	4.2119	4.2594
HGB — 63 leaves, time/risk	4.2442	4.2603	4.2751	4.1790	4.2305

Folds 1–4 each contained 12,927 held-out orders; fold 5 contained 12,926. CV results report the equally weighted mean and sample standard deviation across five folds. Training MAE was calculated on the full training set after fitting. F0 and F1 denote feature sets, not evaluation metrics; ‘original target’ means delivery duration was modelled without a logarithmic transformation.

The historical three-model stack selected among 12 Ridge and six original-target RF candidates using three inner folds within each of the five outer folds. The current experiment replaces RF with the fixed log-target F1 + time/risk configuration and retains the earlier selected Ridge and matched HGB predictions. In both experiments, nonnegative weights summing to one, with zero intercept, minimize MAE on inner out-of-fold predictions. HGB predictions and the auxiliary risk feature are reconstructed within their fitting partitions.

Table A3c. Historical three-model stacking weights using the original-target RF

Outer fold	Ridge	Random forest	HGB
1	0.00%	0.63%	99.37%
2	0.00%	1.69%	98.31%
3	0.00%	2.30%	97.70%
4	0.00%	1.91%	98.09%
5	0.00%	2.44%	97.56%

Historical stacking (Table A3c) achieved outer-CV MAE 4.2374 ± 0.0371 days. Current stacking gives HGB 100% in every fold, matching 4.2378 ± 0.0369 days. Equal averaging yields 4.3969 days for the current three models and 4.2745 for HGB–ExtraTrees. The HGB–ExtraTrees MAE stack reaches 4.2365 days, with 88.76%–94.38% HGB weights; four folds improve and one worsens. The gain is 0.001342 days (0.032%), or 1.93 minutes of mean absolute prediction error, not delivery-time savings. ExtraTrees/HGB residual correlation of 0.974 describes error overlap without proving its causal role. Results in tuned_stacking_comparison.ipynb support retaining HGB within these tested averages and constrained linear stacks; other stacking methods remain untested. No final ensemble refit or test score was produced. The configurations and development folds had already been inspected, so these results do not independently validate the full selection process.

Complete per-fold metrics

Stacking procedure comparisons

A4 Prediction errors and sensitivity analysis

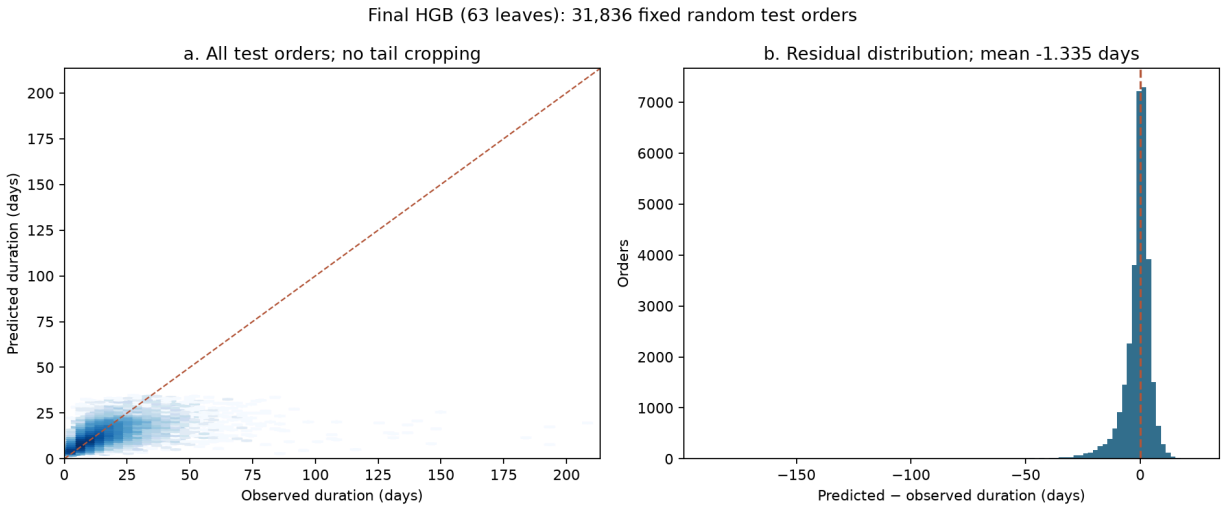


Figure A4a. Predicted versus observed delivery duration (left) and residual distribution (right) for the selected 63-leaf HGB on all 31,836 test orders. Residual equals predicted minus observed duration, in days. All valid extreme durations are retained. Dashed lines indicate perfect prediction (left) and zero residual (right).

Table A4b. Test errors by observed duration and route

Group	n	MAE (days)	Bias (days)
0–7 days	8,584	2.2940	1.9441
7–14 days	13,334	2.6044	0.6293
14–30 days	8,454	5.3738	-4.1787
30–60 days	1,365	18.8525	-18.8266
>60 days	99	66.1881	-66.1881
All sellers same-state	11,431	2.9544	-1.0426

Any seller cross-state	20,405	4.8205	-1.4987

Duration and route are separate grouping dimensions; each covers the same 31,836 orders. Duration intervals after 0–7 days exclude their lower bound and include their upper bound. Observed-duration groups are diagnostic only. Bias is predicted minus observed duration.

The full-test mean signed error was −1.3349 days, the 90th percentile absolute error was 9.0914 days, and 56.93% of predictions were within ±3 days. The data were previously used during development; these results are not an untouched external confirmation.

Removing only the 25 chronology-flagged test orders, without refitting, changed MAE from 4.1505 (n = 31,836) to 4.1523 days (n = 31,811). This does not validate the timestamps or assess retraining without the 59 flagged training orders.

A5 Model interpretation and reproducibility

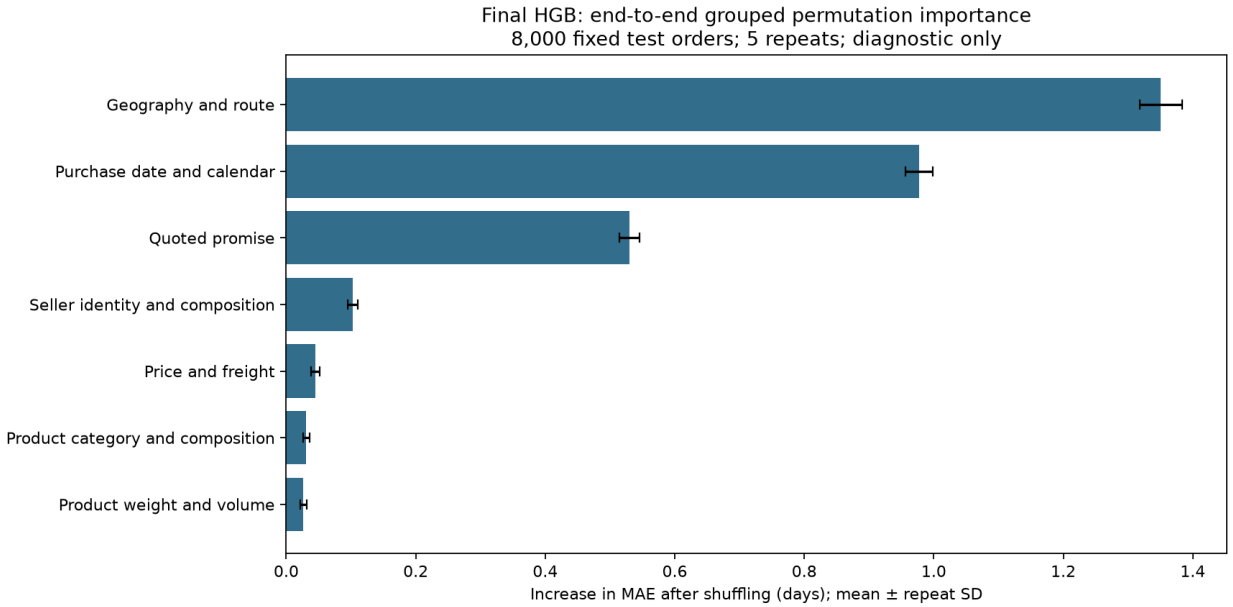


Figure A5a. End-to-end grouped permutation importance for the selected HGB on 8,000 fixed test orders (seed 33). Bars show mean MAE increase over five shuffles per group; error bars show sample SD, not confidence intervals. Derived features and the auxiliary risk score were recomputed without refitting. Increments are not additive or causal effects.

The unshuffled sample MAE was 4.1280 days. Geography/routes, purchase date/calendar and the quoted promise produced the largest mean MAE increases: 1.3504, 0.9771 and 0.5302 days. Correlated groups and unusual shuffled combinations limit interpretation. Date reliance does not establish future-period robustness.

Appendix B. Low-Review Classification

B1 Cohort, split and features

Table B1a. Classification cohort and split

Partition	Orders	Low reviews	Low-review rate
Eligible cohort	97,490	13,796	14.15%
Training	52,254	7,394	14.15%
Validation	13,064	1,849	14.15%
Test	32,172	4,553	14.15%
Pre-T reviews; excluded	426	—	—

Eligibility: item-bearing orders reviewed no earlier than the prediction point T, purchased September 2016–August 2018; 5-fold stratified CV and random seed 5006.

Table B1b: Feature groups: rationale and prediction point

Group (n)	Examples	Rationale	Known at
Order (7)	n_items, n_sellers, total_order_value, *freight_ratio*	Split shipments rated once; value and freight share set expectations (Zaghloul et al., 2024)	order approval
Product (6)	weight, volume, main_category	Carrier handling class; category-specific complaint rates	order approval
Geography (9)	max_distance_km, same_state, regions, customer_zip2	Transit exposure; last-mile variation (Vakulenko et al., 2019)	order approval
Timing (8)	promised_days, approval_lag_hours, calendar, black_friday_week	The promise is the expectation judged against (Cui et al., 2024); demand peaks	order approval
Payment (4)	payment_total*, instalments, type	Value sensitivity	order approval
Seller history (6)	prior orders, tenure, late rate, low-review rate, seller_id_enc	Seller management's warning signal, point-in-time from training-row outcomes	purchase time
Route history (3)	route prior orders, delivery days, late rate	Same for the state-to-state lane	purchase time

Delivery status at T (9)	is_delivered, wd_actual_delivery_time, *wd_delivery_time_delta*, carrier hand-over / transit, delivery_vs_seller_prior	Lateness drives low ratings (φ = 0.43); working-day versions from Zaghloul et al.; missing with indicator for the 9.9% undelivered at T	survey trigger T

* the five literature features. Source: config/feature_roles.csv.

B2 Model configurations and selection

Table B2a. Selected tuning settings and protocol

Model	Selected tuning settings
Logistic Regression	C = 0.1
Decision Tree	max_depth = 9; min_samples_leaf = 138
Random Forest	478 trees; depth = 21; min leaf = 70; max_features = 0.356
CatBoost	depth = 4; 1,000 iterations; learning rate = 0.0318
Protocol	5-fold stratified CV; seed 5006; fold-fitted histories; validation threshold

Table B2b. Five-fold CV and validation PR-AUC

Candidate	Five-fold CV PR-AUC ± SD	Validation PR-AUC
Logistic Regression	0.4986 ± 0.0133	0.4687
Decision Tree	0.4897 ± 0.0084	0.4691
Random Forest (selected)	0.5147 ± 0.0104	0.4913
CatBoost	0.5070 ± 0.0101	0.4834
Stacking	—	0.4892

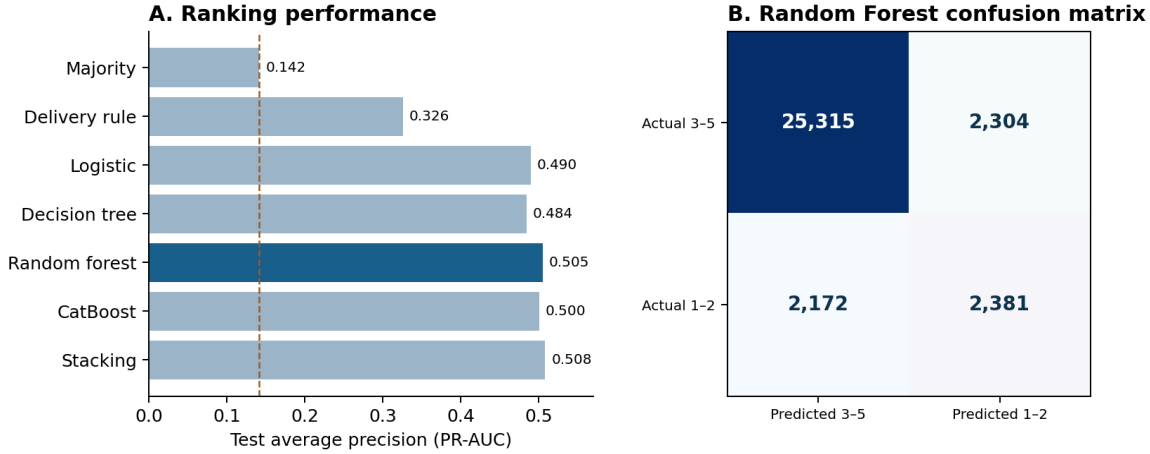
B3 Imbalance and calibration sensitivity

Table B3a. Random Forest imbalance/calibration sensitivity

Strategy	Validation PR-AUC	Test PR-AUC	Test Brier
Unweighted	0.4914	0.5058	0.0923
Class weights (reported model)	0.4913	0.5047	0.1453
Random oversampling	0.4892	0.5055	0.1457
SMOTE	0.4773	0.4976	0.0991
ADASYN	0.4767	0.4964	0.0988
Class weights + isotonic calibration (validation)	0.4832	0.4875	0.0927

Random forest imbalance sensitivity. Same hyper-parameters in every row; resampling inside the Pipeline; thresholds frozen on validation. The class-weighted row is the model reported throughout Sections 3–4; the other rows are refits for comparison.

Figure B3b. Test PR-AUC by model and selected Random Forest confusion matrix (4,553 positive cases).



B4 Operating points

Table B4a. Operating points for the selected 478-tree Random Forest on the test set (4,553 low reviews; 27,619 other reviews). Thresholds are derived from validation scores.

Policy (test %)	Cutoff	Recall (TPR)	FPR	Satisfied customers contacted	Youden J
Top 5% (4.77%)	0.8858	0.239	0.016	448	0.222
Top 10% (10.31%)	0.7661	0.435	0.048	1,338	0.386
F1 optimum (14.56%)	0.5896	0.523	0.083	2,304	0.440
Top 20% (20.47%)	0.4706	0.605	0.139	3,831	0.466
Top 25% (25.87%)	0.4018	0.655	0.193	5,338	0.462
Top 30% (31.24%)	0.3650	0.695	0.249	6,884	0.446
Top 50% (51.09%)	0.2784	0.812	0.461	12,737	0.351

Capacity policies use validation-score quantiles; the F1 row uses the saved validation optimum. Each cutoff is applied unchanged to test predictions. Parentheses report the actual test contact share, which can differ from the validation quota. Capacity comparisons are retrospective descriptions, not an additional model-selection step. FPR measures satisfied customers contacted unnecessarily; every 1% corresponds to about 276 test orders.

B5 validation permutation importance

Figure B5a. Selected Random Forest validation permutation importance: decrease in PR-AUC.

